

Safety · Accuracy · Observability

An Indian e-commerce support bot launched without evals or guardrails. Within two days, WhatsApp screenshots of it issuing ₹50,000 refunds on demand went viral. The company paid out ₹2.1 crore before pulling it offline. The fix was one output check.

Which LLM in your org today is one prompt injection away from a...

WHAT'S INSIDE

- 01 Three layers of evaluation
- 02 Guardrails — input + output
- 03 Prompt injection — the 2026 attack
- 04 Observability in production
- 05 Analogy

WHY THIS LESSON MATTERS

₹2.1 crore. 48 hours. One missing guardrail.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

Distinguish offline / online / judge-based eval.

2

Golden set + eval harness.

3

Input + output guardrails.

4

Detect + prevent prompt injection.

5

Log + monitor LLM systems in production.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

Three layers of evaluation

Key concept of this lesson.

02

Guardrails — input + output

Deterministic checks around the LLM

03

Prompt injection — the 2026...

Direct + indirect

04

Observability in production

Log, alert, review

01

PART 1 OF 4

Three layers of evaluation

Key concept of this lesson.

Three layers of evaluation

A core idea you'll use throughout this lesson.

DEFINITION

Three layers of evaluation

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

Guardrails — input + output

Input: PII redaction, toxicity, off-topic, injection detection.

Output: factuality check, schema validation, safety classifier.

NOT part of the LLM — separate code.

Fail fast, fail loud.

BOUND

Around LLM

Regex, classifiers, business rules.

KEY POINTS

- Input: PII redaction, toxicity, off-topic, injection detection
- Output: factuality check, schema validation, safety classifier
- NOT part of the LLM — separate code
- Fail fast, fail loud

Prompt injection — the 2026 attack

Direct: hostile user input overrides system prompt.

Indirect: hostile content hidden in retrieved docs.

Defence 1: input sanitisation + intent classifier.

Defence 2: strict output schema + post-validation.

Defence 3: least-privilege tools (no destructive ops).

Defence 4: human-in-the-loop for sensitive actions.

CLASS

Defining attack

Of the 2026 AI-production landscape.

DEFINITION

Prompt injection — the 2026 attack

Direct + indirect

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

Observability in production

Log (prompt, completion, tokens, latency, cost) per call.

Monitor p95 latency + hallucination rate.

Alert on drift; accuracy decay triggers review.

Weekly sample review — real conversations.

HABITS

Log · Alert · Review

Weekly, not annually.

KEY POINTS

- Log (prompt, completion, tokens, latency, cost) per call
- Monitor p95 latency + hallucination rate
- Alert on drift; accuracy decay triggers review
- Weekly sample review — real conversations

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Pause and reflect

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1

2

3

4

Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Where will you apply this in your next project?

Q2

What was the single biggest 'aha' from this lesson?

Q3

Which habit will you start using this week?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

Indirect prompt injection:

A

None

B

Through retrieved docs

C

Via UI

D

Via DNS

Indirect prompt injection:



CORRECT ANSWER

Through retrieved docs

WHY

Attacker embeds instructions in a doc the agent reads

Best defence against injection (one of many):

A

Bigger model

B

Least-privilege tools

C

Smaller prompts

D

More temperature

Best defence against injection (one of many):

CORRECT ANSWER



Least-privilege tools

WHY

Least-privilege + schema + HITL + classifier, layered

Must log per LLM call:

A

Just result

B

Prompt + completion + tokens + latency + cost

C

Nothing

D

Memory

Must log per LLM call:



CORRECT ANSWER

Prompt + completion + tokens + latency + cost

WHY

Enough to debug, monitor drift, and compute cost

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

A team applies the lesson's main idea to a real workflow — and the results show up within a quarter.

THE TAKEAWAY

The right tool, applied to the right problem, beats the loudest idea every time.

WHAT TO NOTICE

1

Three layers of evaluation

2

Guardrails — input + output

3

Prompt injection — the 2026...

4

Observability in production

Guardrails are cheaper than outages.

"Every production LLM system without evals + guardrails is a liability waiting to happen."

DELIVERABLES

- NEXT
- Projects Saturday — ship a RAG + an agent.
- Document one decision you made because of this lesson.
- Document one decision you made because of this lesson.

WHAT 'GOOD' LOOKS LIKE

ONE THING TO REMEMBER

“

Knowing the right tool is half the battle. Doing the small thing today is the other half.

— AI Learning Hub

END OF LESSON 5

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.